

3-Strategy Portfolio

Backtest & Walk-Forward Validation Report

Alpha TA (40%) | Cup & Handle (35%) | XS Mean Reversion (25%)

Walk-Forward: Train 2020-01 to 2023-12 | Test 2024-01 to present

Fees: Taker 0.035%/side + Slippage 0.05%/side = 0.085%/side

Sizing Base: 40,000 (4x on 10,000 equity)

Exchange: Hyperliquid | Cadence: 4h (all strategies)

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Walk-Forward Summary

TRAIN (2020-2023)

TEST / OOS (2024+)

CAGR +7.52%

Sharpe 1.68

Sortino 3.04

Calmar 2.49

Max DD -3.02%

Total Return +26.75%

Years 3.3

CAGR +7.22%

Sharpe 1.56

Sortino 2.98

Calmar 1.93

Max DD -3.75%

Total Return +19.39%

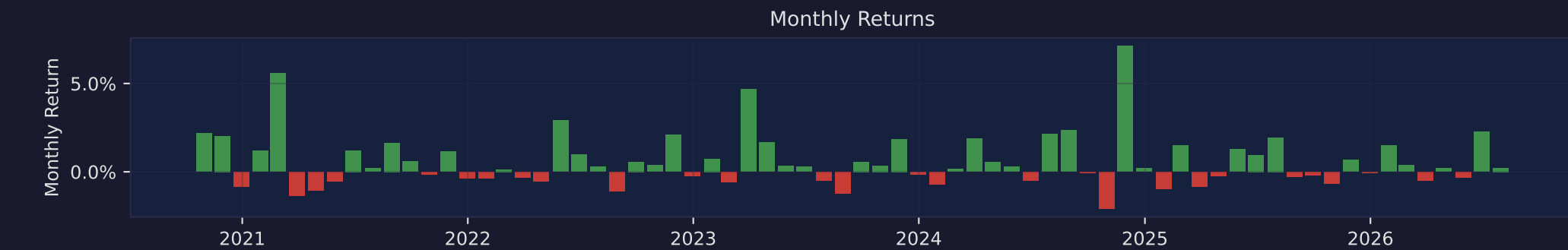
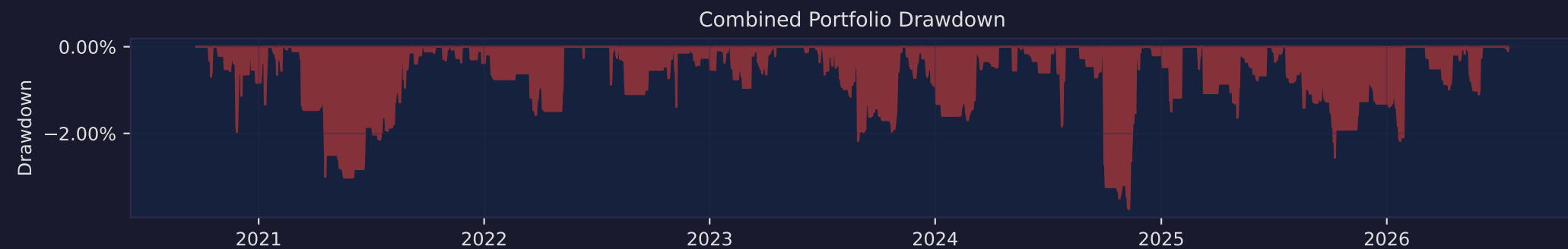
Years 2.5

Weights: ATA 40% / C&H 35% / XSMR 25% | Sharpe decay: +7% | Verdict: PASS

Train vs Test (OOS) — Per-Strategy Metrics

Strategy	Period	CAGR	MDD	Sharpe	Sortino	Calmar
Alpha TA	Train	+7.5%	-5.8%	1.06	2.19	1.29
	Test	+8.9%	-5.8%	1.19	2.26	1.54
C&H 30-coin	Train	+3.9%	-4.8%	0.77	1.47	0.81
	Test	+3.3%	-5.3%	0.53	1.03	0.62
XS MR	Train	+12.0%	-7.7%	1.13	1.89	1.55
	Test	+9.5%	-5.1%	1.32	2.76	1.88
Combined 40/35/25	Train	+7.5%	-3.0%	1.68	3.04	2.49
	Test	+7.2%	-3.8%	1.56	2.98	1.93

Equity Curves & Drawdowns



Individual Strategy Analysis

Alpha TA



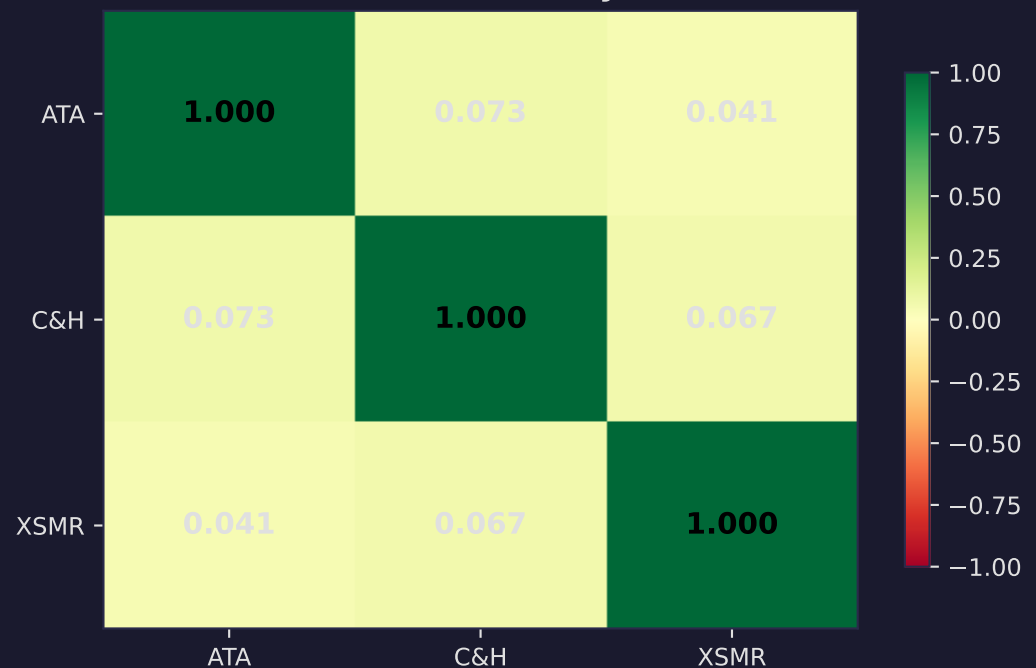
Cup & Handle



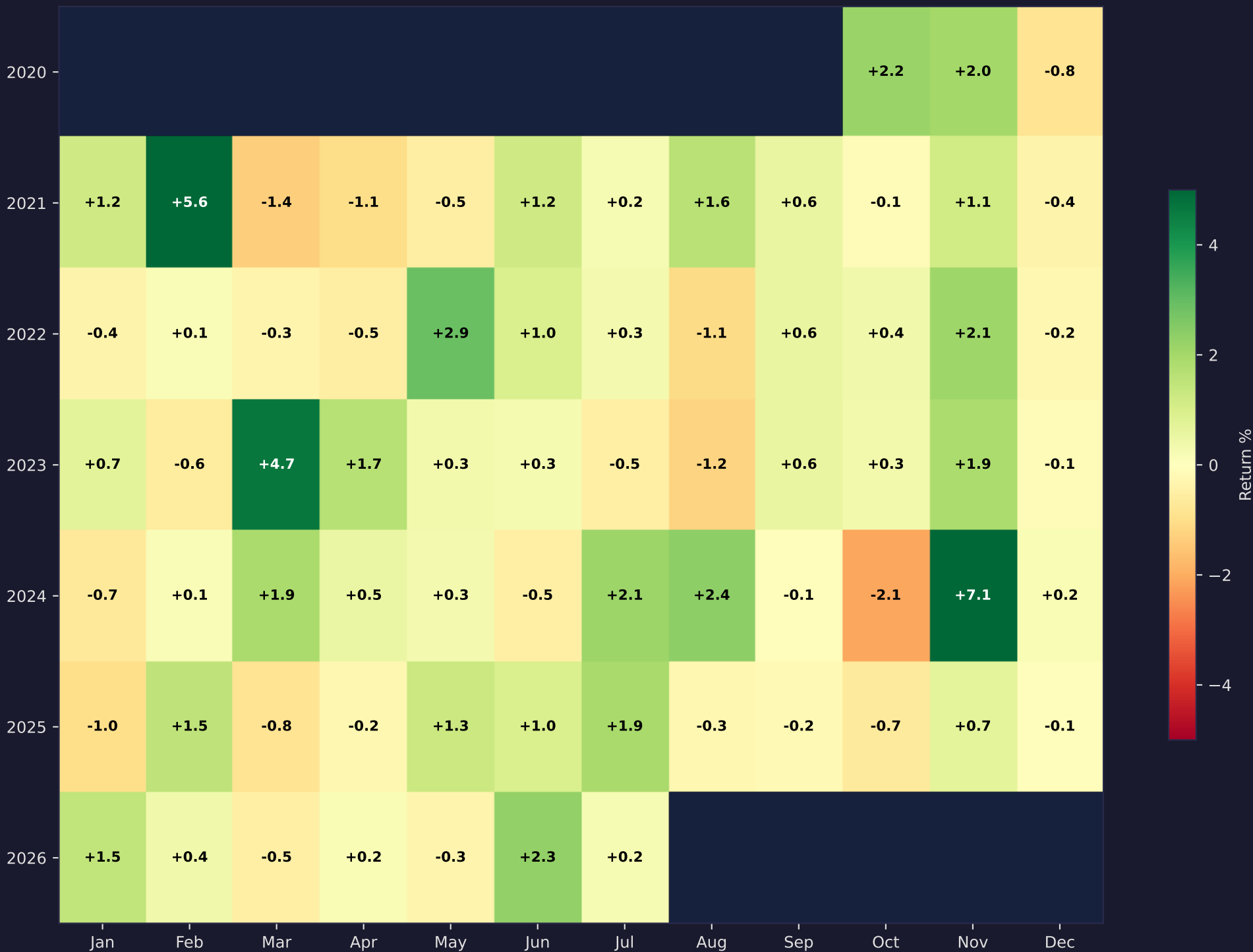
XS Mean Reversion



Correlation Matrix (Daily Returns)



Monthly Returns Heatmap (Combined Portfolio)

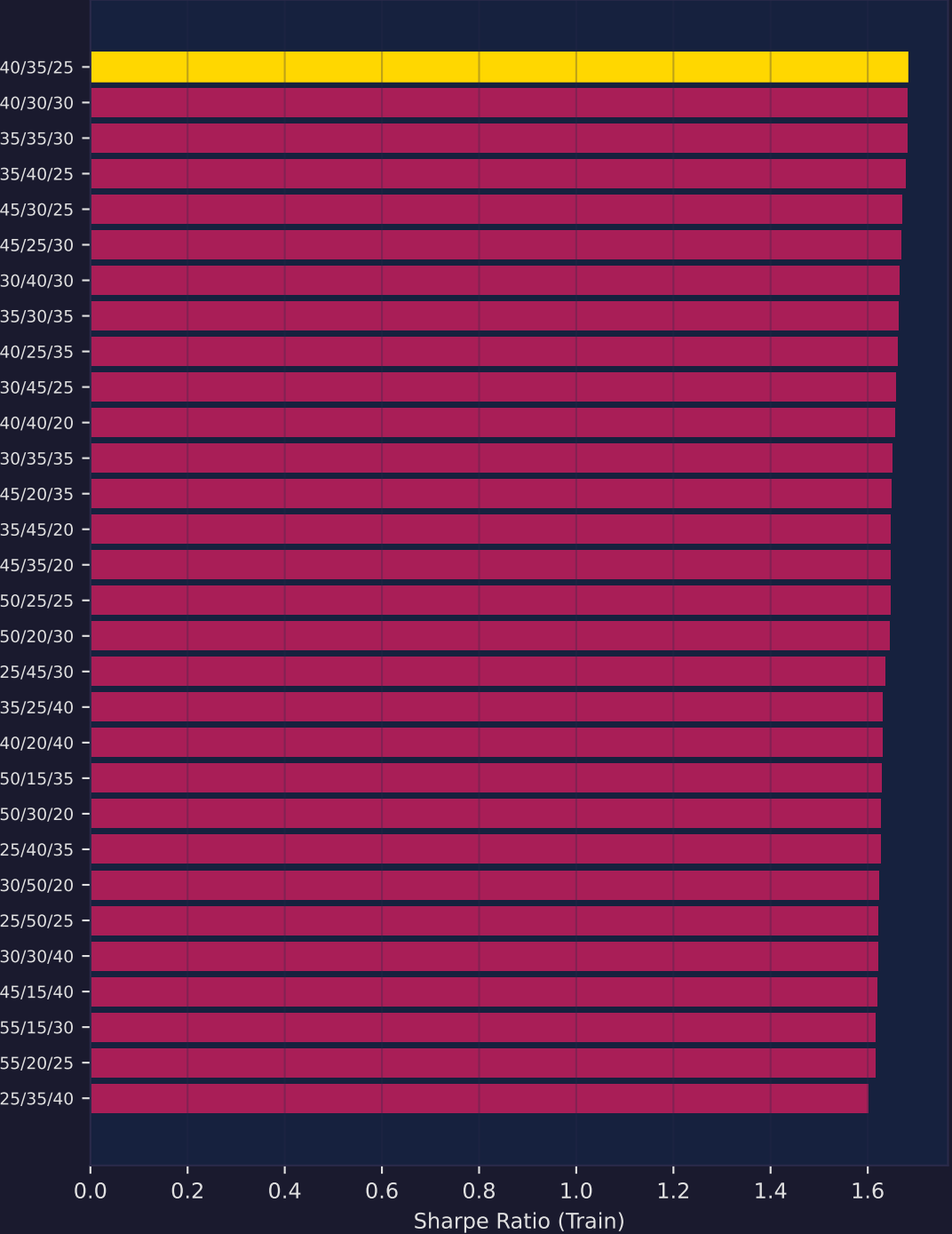


Yearly Returns Breakdown

Year	ATA	C&H	XSMR	Combined	Period
2020	+6.7%	+0.8%	+1.3%	+3.4%	TRAIN
2021	+0.4%	+6.6%	+23.6%	+8.2%	TRAIN
2022	+5.4%	+4.7%	+3.3%	+4.8%	TRAIN
2023	+12.3%	+0.7%	+11.9%	+8.1%	TRAIN
2024	+11.0%	+6.0%	+20.1%	+11.6%	TEST
2025	+2.1%	+3.4%	+3.9%	+3.1%	TEST
2026	+9.6%	-0.9%	+1.0%	+3.7%	TEST

Weight Optimization (Train Period Only)

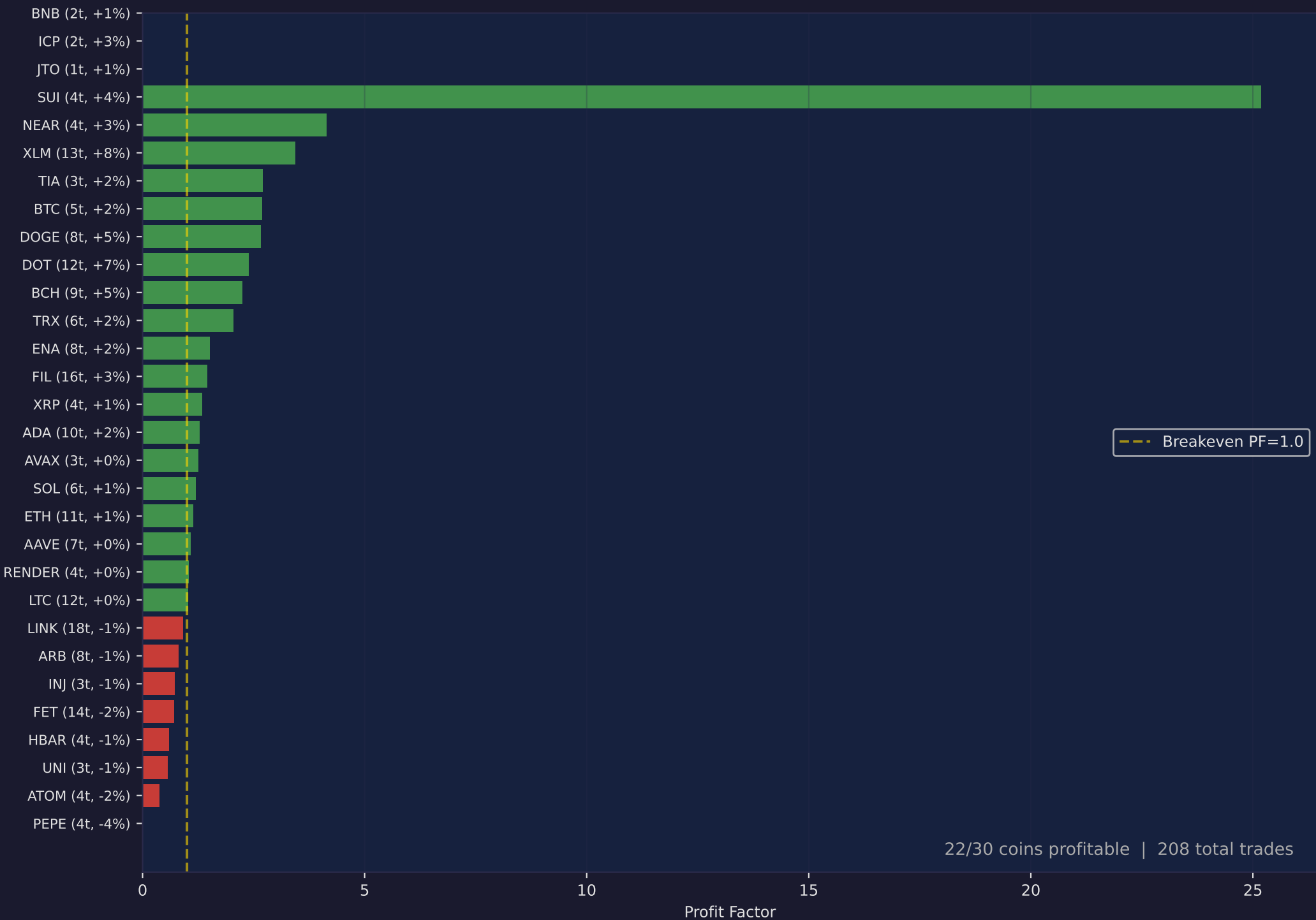
Top 30 Weight Combos by Sharpe



Top 10 Detail

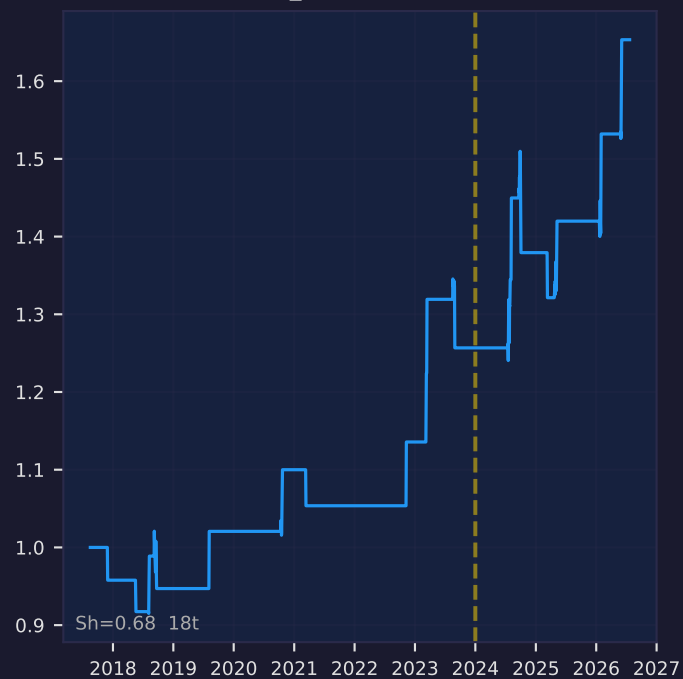
#	ATA	C&H	XS	Sharpe	CAGR	MDD
1	40%	35%	25%	1.68	+7.5%	-3.0%
2	40%	30%	30%	1.68	+7.9%	-3.3%
3	35%	35%	30%	1.68	+7.7%	-3.2%
4	35%	40%	25%	1.68	+7.3%	-2.8%
5	45%	30%	25%	1.67	+7.7%	-3.3%
6	45%	25%	30%	1.67	+8.1%	-3.5%
7	30%	40%	30%	1.67	+7.6%	-3.0%
8	35%	30%	35%	1.66	+8.2%	-3.5%
9	40%	25%	35%	1.66	+8.3%	-3.7%
10	30%	45%	25%	1.66	+7.1%	-2.7%

Cup & Handle — Per-Coin Breakdown (30-Coin Universe)

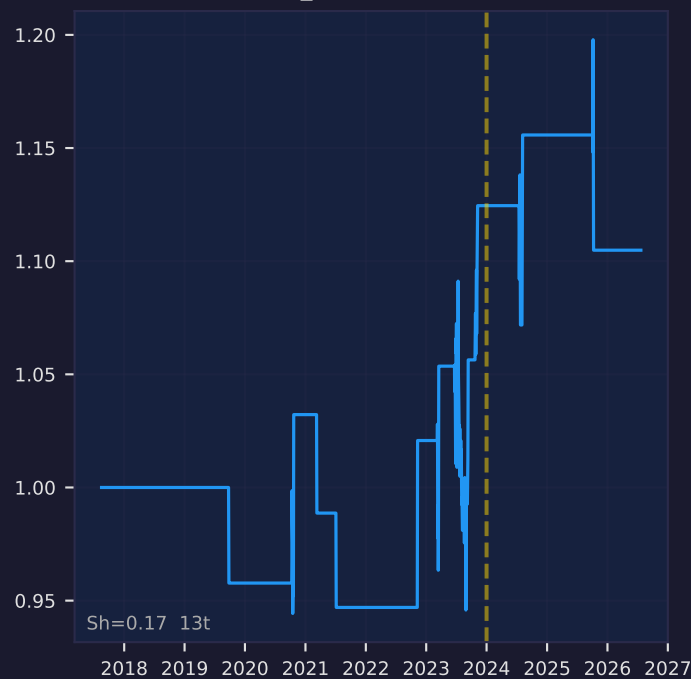


Alpha TA — Per-Sleeve Equity Curves

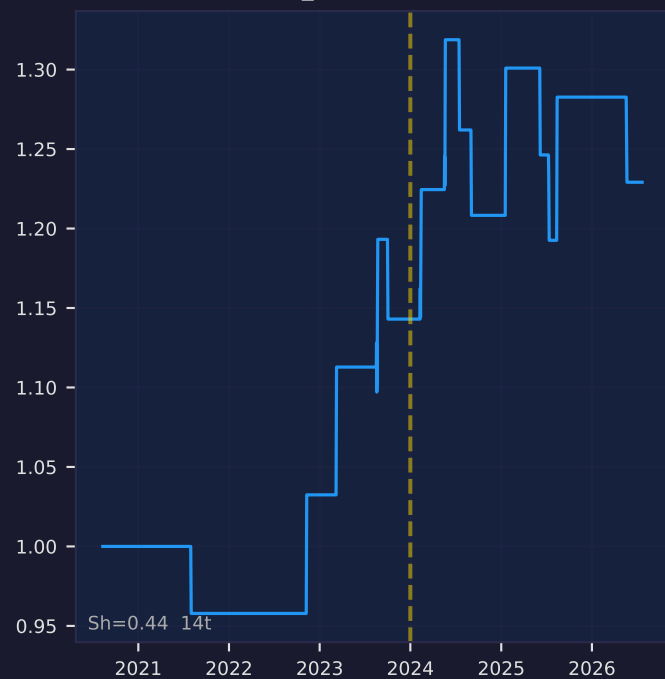
AA_BTC (63.7%)



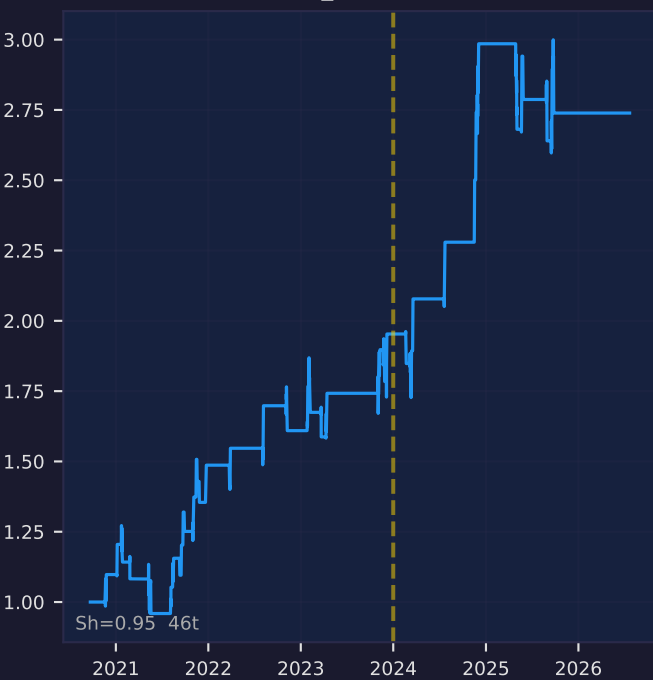
AA_ETH (10.6%)



AA_SOL (10.6%)



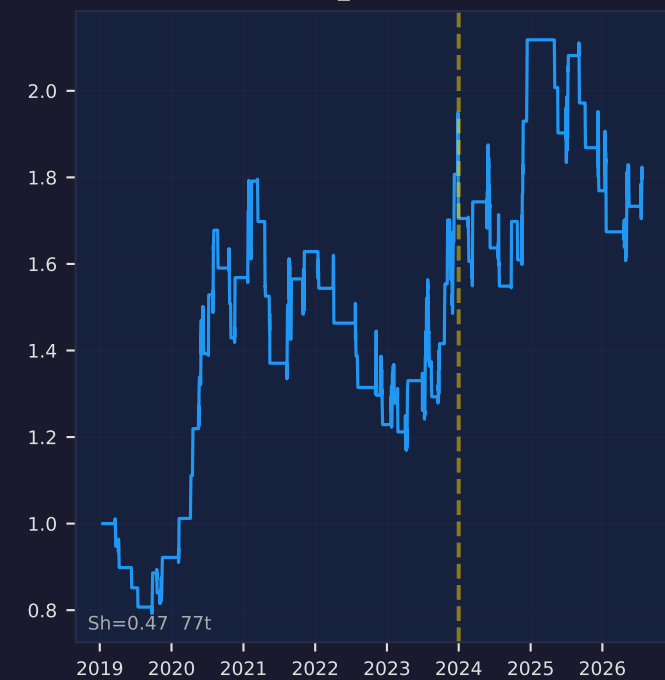
EMAREJ_AVAX (5.0%)



EMAREJ_ETH (5.0%)



EMAREJ_LINK (5.0%)



Projected Returns @ \$40,000 Base (4x Leverage)

Full Period

CAGR	+7.38%	\$+2,954/yr	+29.5% on equity
Monthly avg	+0.60%	\$+242/mo	
Monthly median	+0.28%	\$+113/mo	
Best month	+7.12%	\$+2,846	
Worst month	-2.09%	\$-837	
Max DD	-3.75%	\$-1,500	-15.0% of equity
Sharpe	1.63		
% months positive	60%		

OOS Only (2024+)

CAGR	+7.22%	\$+2,890/yr	+28.9% on equity
Monthly avg	+0.63%	\$+251/mo	
Monthly median	+0.21%	\$+83/mo	
Best month	+7.12%	\$+2,846	
Worst month	-2.09%	\$-837	
Max DD	-3.75%	\$-1,500	-15.0% of equity
Sharpe	1.56		
% months positive	60%		

Strategy Configuration Details

ALPHA TA (40% allocation)

OPA 85%: BTC 75% / ETH 12.5% / SOL 12.5% (daily, SL 4%, TP 8%)

EMArej 15%: AVAX / ETH / LINK (4h, SL 5%, TP 10%, ADX>25 + CCI filter)

Long + short signals, funding rate adjusted

CUP & HANDLE (35% allocation)

Universe: ALL 30 coins (no cherry-picking)

4h timeframe, volume-filtered (1.5x SMA50)

Risk-based sizing: 1% of C&H capital per trade, sized by stop distance

Total backtest trades: 208

Win rate: 49%

XS MEAN REVERSION (25% allocation)

Pool: ADA, ATOM, BNB, DOGE, ETH, LINK, LTC, TRX, XLM, XRP (anchored 2020, delisted EOS removed)

Daily, long-only, 3 slots per rebalance (5-day cycle)

Z-score entry < -1.7, early exit at $|z| < 0.3$ (causal: yesterday's z, exit at today's open)

USDT.D 50-EMA regime gate: blocks longs when dominance > EMA

RISK MANAGEMENT

Sizing base: 40,000(4x on 10,000 equity)

Virtual sub-accounting: each strategy tracks own positions

Net position submitted to exchange (13 coins overlap across strategies)

SL/TP: USD-weighted average across active sleeves; XSMR has signal-managed exits

Step-up plan: 4x now -> 6x after 2 months -> 8x after full quarter